

A
Project
Report On
“IoT Based Gas Leakage Detection System”

Submitted to the
Savitribai Phule Pune University
In partial fulfillment for the award of the IoT Mini Project
in
Artificial Intelligence & Data Science Engineering
by

Sr. No.	Name of Student	Roll No.
1)	Mohan S. Ugale	66
2)	Jayesh B. Pendharkar	51
3)	Prathamesh K. Khule	30
4)	Shivam A. Chavan	09

Under the guidance of

Prof. Mrs. M. P. Joshi



Department Of Artificial Intelligence & Data Science Engineering
PVG's College of Engineering & S. S. Dhamankar Institute of
Management, Nashik - 422004
2023-2024



Department of AI&DS Engineering
Pune Vidyarthi Griha's College of Engineering & S.S.
Dhamankar Institute of Management, Nashik

CERTIFICATE

This is to certify that:

Name of Student	Class	Roll No.
1. Mohan Suresh Ugale.	SE	66
2. Jayesh Bhikaji Pendharkar.	SE	51
3. Prathamesh Kiran Khule.	SE	30
4. Shivam Annasaheb Chavan.	SE	09

From the institute - Pune Vidyarthi Griha's College of Engineering & S.S. Dhamankar Institute of Management, Nashik has completed the Project (Mini Project in IoT) having title “**IoT Based Gas Leakage Detection System**” during the Academic Year 2023-24 under the supervision and guidance of Professor **Mrs. M. P. Joshi** in partial fulfillment of the requirement for **SE (Artificial Intelligence & Data Science Engineering) – 2019 course** of Savitribai Phule Pune University, Pune.

Date: 30/04/2024

Place: Nashik

Prof. Mrs. M. P. Joshi
Guide

Prof. Mr. S. G. Chordiya
Head of the Department



PVG's College of Engineering & S. S. Dhamankar
Institute of Management, Nashik - 422004
DEPARTMENT OF AI&DS ENGINEERING

Institute Code: 4086	Academic Year: 2023-24	Program: AI & DS
Course: Internet of Things (IoT).		Course Code: 217531
Semester: 4th	Class: Second Year Engineering.	Date of Report: 30/04/2024

Title of Mini-Project: IoT Based Gas Leakage Detection System.

1.0 Problem Definition: IoT Based Gas Leakage Detection System aims to develop an IoT-based Gas Leakage Detection System using Arduino and an MQ2 sensor. This system detects the presence of gas in the environment and provides alerts to prevent potential hazards.

2.0 Rationale: Gas leaks can pose significant risks such as fire hazards and health issues. An IoT-based Gas Leakage Detection System offers a proactive approach to monitor gas levels and provide timely alerts, enhancing safety in various environments. An IoT-based Gas Leakage Detection System is crucial for ensuring safety in environments where gas leaks can pose serious risks. Here's why this system is essential:

Safety: Gas leaks can lead to fire hazards and health risks. Detecting gas leaks promptly can prevent accidents and protect lives.

Automation: By integrating IoT technology, the system can monitor gas levels continuously and provide real-time alerts without human intervention.

Efficiency: Automated gas detection enhances operational efficiency by quickly identifying and addressing potential hazards.

3.0 Aim /Benefits of Mini-Project:

The aim of this mini-project is to design and implement a cost-effective and efficient gas leakage detection system. The benefits include:

- Early detection of gas leaks.
- Prevention of potential hazards.
- Real-time monitoring capabilities.
- Cost-effective solution using Arduino and MQ2 sensor.

4.0 Literature Review:

Gas leakage poses serious safety risks in various environments, including residential, commercial, and industrial settings. Traditional gas detection methods often rely on manual inspection or standalone sensors, which may not provide timely alerts or comprehensive monitoring. IoT-based Gas Leakage Detection Systems have emerged as a promising solution to address these challenges. By integrating sensors with Internet connectivity and data processing capabilities, these systems offer real-time monitoring and alerts for gas leaks.

5.0 Actual Methodology followed:

- a) Finalizing the project topic i.e. **IoT Based Gas Leakage Detection System.**
- b) Distributed the work among team members.
- c) Interpret and Filter the information collected.
- d) Collection of required sensors and other hardware.
- e) Actual Implementation of Project.
- f) Prepared the Project report.
- g) Submitted the report to Subject Teacher.
- h) Submitted the whole project (hard, soft copy) along with the viva.

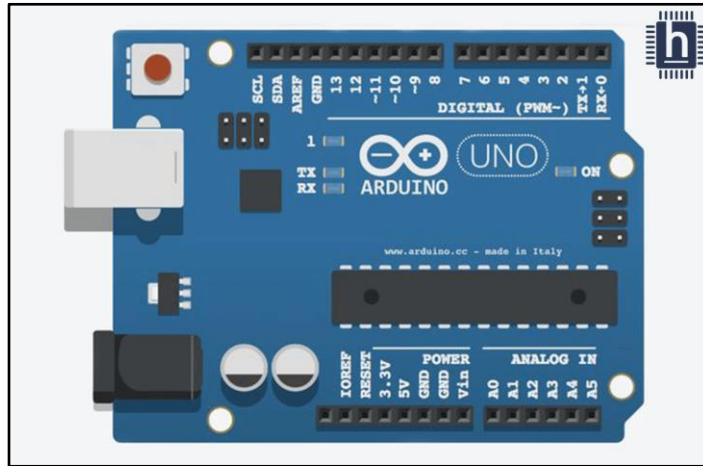
6.0 Actual Resources used:

Sr. No.	Name of Resource/material	Specifications	Qty
1	Computer system	Hp intel i5 Processor, 8 GB Ram. 512 GB SSD	01
2	Arduino Board	Arduino UNO	01
3	Data Cable	USB Cable Type A to B	01
4	Gas Sensor	MQ2	01
5	LCD (Display)	LCD 16×2	01
6	LEDs	Green, Red	02
7	Jumper Wires	Male- Female, Male- Male	10,15
8	LPG Gas Lighter	Filled with LPG Gas	01
9	Operating System	Windows 11	01
10	Software	Arduino IDE, Chrome, Microsoft Word	03

7.0 Components Used In Project:

7.0.1. Arduino UNO Circuit:

The Arduino Uno circuit is the fundamental foundation of the Arduino platform, renowned for its versatility and ease of use in electronics projects. At the core of the Arduino Uno lies the ATmega328P microcontroller, a key component responsible for executing program instructions and interfacing with external devices. The circuit also incorporates a crystal oscillator operating at 16 MHz, providing the necessary clock signal for precise timing operations. Additionally, a voltage regulator, typically the 7805, ensures a stable 5V supply voltage for the microcontroller and other connected components. The USB connector enables easy connectivity to a computer for programming and power supply purposes. Alongside, the Arduino Uno features an array of digital and analog input/output pins, allowing seamless interfacing with various sensors, actuators, and peripherals. Furthermore, power pins, including 5V, 3.3V, and GND connections, facilitate powering external devices and ensuring proper grounding. The presence of LEDs, such as power, TX, and RX indicators, offers visual feedback on the Arduino's operational status and communication activities. Lastly, the reset button provides a means to manually restart the microcontroller, while the ICSP header enables programming of the ATmega328P using an external programmer like the AVRISP mkII. Together, these components form a robust and versatile circuit layout, empowering enthusiasts and professionals alike to bring their electronic creations to life with Arduino Uno.



Arduino UNO Circuit Board

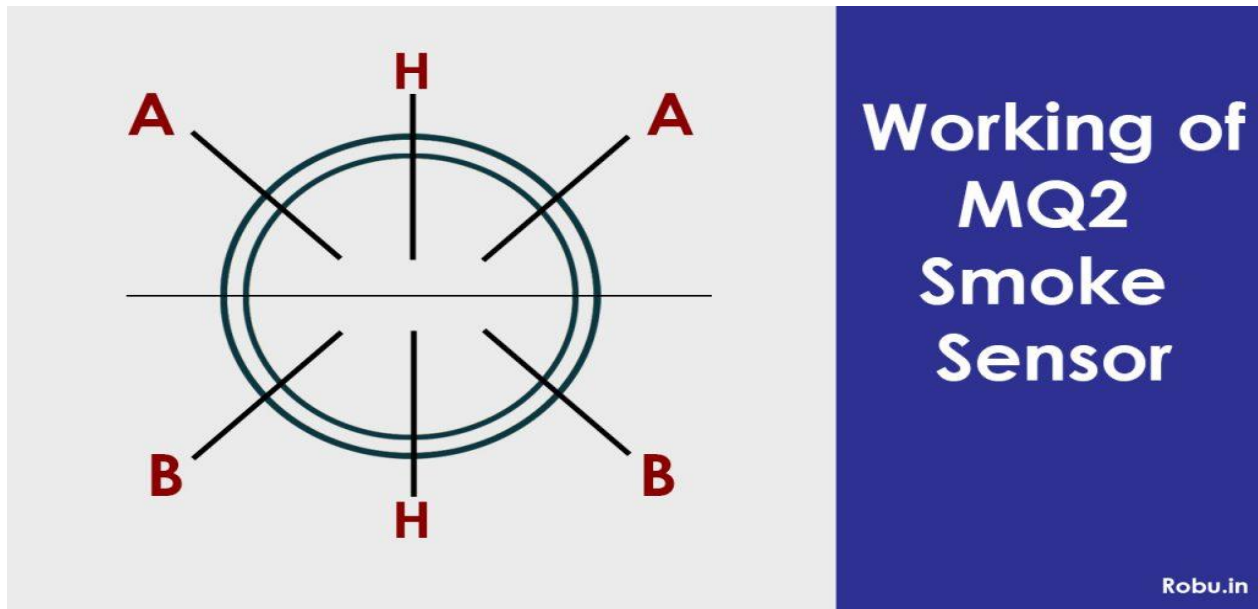
7.0.2. MQ2 Sensor:

MQ2 is a Gas sensor Module that can be used to sense gases like LPG (Cooking Gas), Smoke, Alcohol, Hydrogen, Methane, and Carbon Monoxide concentrations in air. This type of sensor is suitable in making projects like Air Quality monitoring systems, fire alarms, Gas Leakage detection, Alcohol detection-based Ignition system in vehicles, combining the same with IoT, the applications are infinite. The MQ2 sensor is a Metal Oxide Semiconductor (MOS) based sensor which means just like a Light-dependent resistor (LDR), the MQ2 is equipped with a ChemiResistor (stannic oxide) which means the resistance of the sensing material changes when an appropriate gas comes in contact with the same. Depending on the level of concentration of the gas, the sensor outputs a potential difference, which changes the resistance of the material inside the sensor. This change is used to determine the type of gas as well as the concentration of its presence in the atmosphere in ppm (parts per million) from the Analog signal from the sensor (Analog out pin) while Digital values (Digital out pin) from the sensor can be used to detect the presence of these gases.

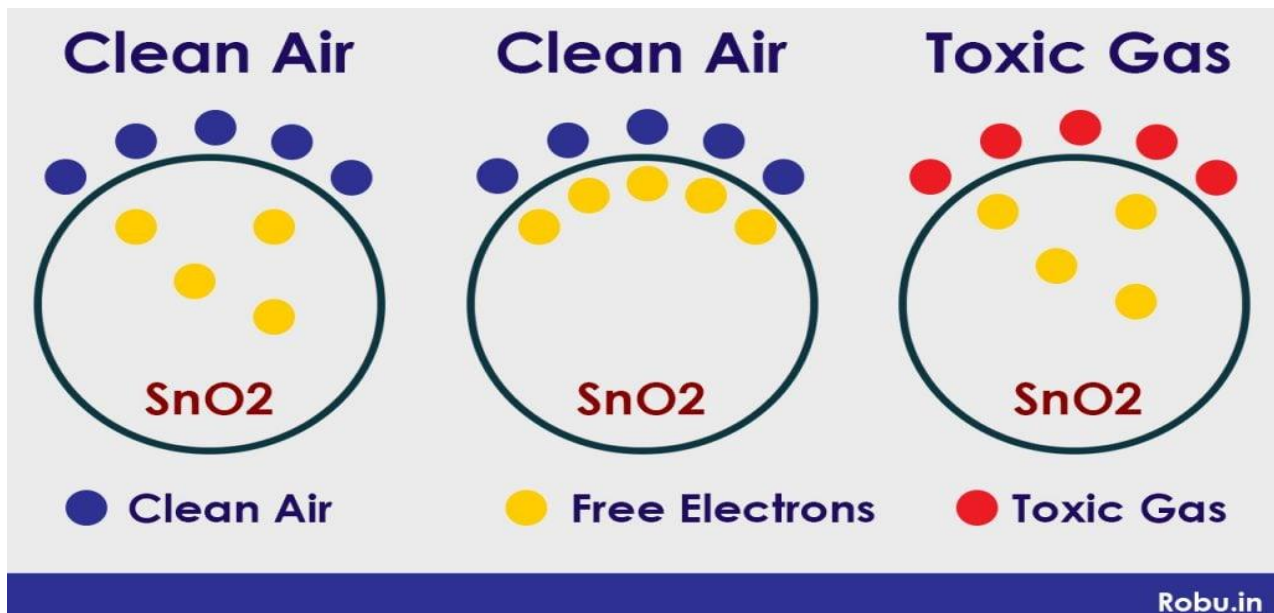


MQ2 Sensor

How Does Gas Sensor Work?

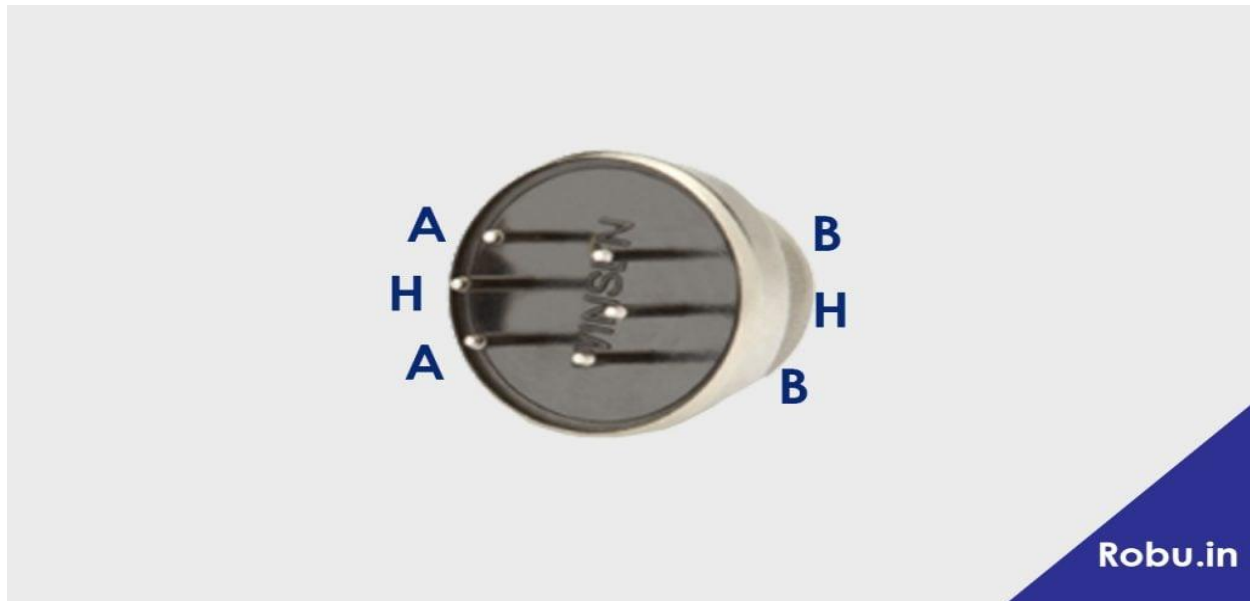


The sensor has a sensing element in order to sense the smoke or any gaseous material. In this sensor, the sensing material is a stainless steel mesh. This mesh contains aluminium-oxide-based ceramic, coated with tin dioxide. There are mainly 6 leads in the sensing element. Two are used for heating of sensing elements, the remaining four are used for the output signal. Let's consider that we want to identify the Oxygen gas, the oxygen atoms get absorbed on the sensing element when it will be heated at a higher temperature.



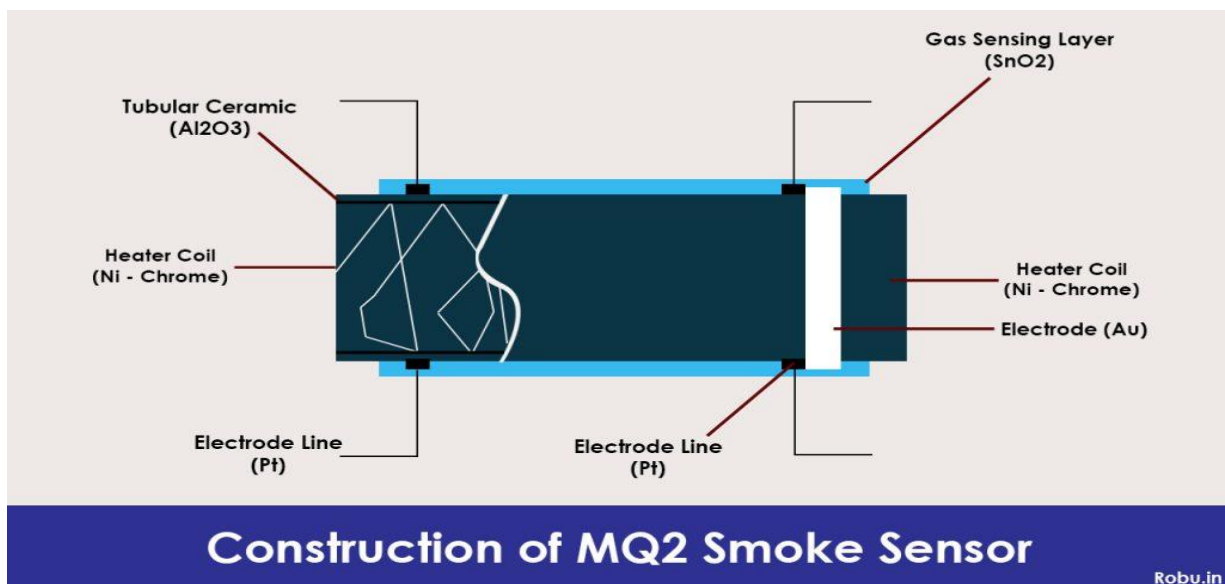
Tin oxide has donor electrons present in it, which are attracted towards oxygen. So by identifying the donor electrons, we can detect the amount of oxygen that gets absorbed as well as we can oppose the flow of current to give the signal. When reducing gases are present on the sensor, oxygen atoms try to interact with reducing gases and reduce the absorbed surface density of the oxygen atom. Hence current can flow through the sensor to generate an analog output signal. This output signal generates the voltage values by which we will get an idea about the concentration of gases. When the concentration of gas is high voltage will be high.

What is MQ2 Gas Sensor?



The MQ2 module of the gas sensor is useful for gas leakage detection in the home as well as in the industries. It observes the level of gases present in the atmosphere. The gas sensor operates on 5V and 800mW. Depending on the level of concentration, the sensor observes the potential difference and this changes the resistance of the material inside the sensor. The output voltage value gives us the type of gas. Normally these sensors are available as a comparator. These comparators are able to set up a threshold value for gas concentration. The digital pin will go high only when the concentration of the gas is above the threshold value. The sensor will work in analog mode to detect the concentration of the gas.

Construction of the Gas Sensor



Generally, all types of gas sensors including MQ2 gas sensors are constructed by metal oxide semiconductors. These gas sensors have one sensing element which consists of the following elements.

- Tubular ceramic

- Electrodes
- Heater Coil
- Gas sensing layer
- Electrode line

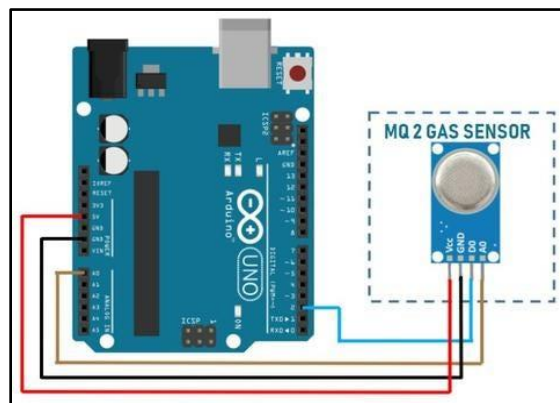
This sensing element is responsible to process all the parameters and detect the type of gas.

How to Adjust the Sensitivity of Gas Sensor

In order to adjust the sensitivity of this sensor for the particular type of gas, you need to keep this sensor in the particular gas type you want to detect. Now we have to rotate the potentiometer on the sensor. Rotate it till the Red LED on the sensor gets ON. If we have to increase the sensitivity, rotate it clockwise, and if we have to decrease the sensitivity we have to rotate it anticlockwise. When we are rotating this potentiometer basically we are adjusting the threshold value of the sensor. The comparator on this board is constantly checking the threshold value. Once this threshold gets crossed the digital pin goes high and the LED will turn on. In this way calibrate your sensor, it may take some time but once you calibrate it, it will give you accurate readings.

Interfacing MQ2 Sensor with Arduino UNO

Arduino	MQ-2 Gas/Smoke Sensor
Analog Pin	A0
Digital Pin	D0
GND	GND
5V	VCC



7.0.3. LCD I2C Display:

LCD Display:

LCD stands for Liquid Crystal Display. LCD is a flat-paneled display. It uses liquid crystals combined with polarized to display the content. LCD uses the light modulation property of LCD. LCD is available both in Monochrome and Multicolor. It cannot emit light directly without a backlight. In some LCDs, It displays the content only with the help of a backlight in a dark place.

I2C communication:

I2C or IIC stands for Inter-Integrated Communication. I2C is a serial communication interface to communicate with other I2C devices. I2C uses multi-master / multi slave method. I2C uses 2 lines named SCL and SDA for transmission/reception and another 2 lines for power supply and ground. Each and every I2C device has I2C address to identify. I2C addresses of multiple devices may have the same address. The address is in the format of “0x20” (Example address).

I2C LCD:

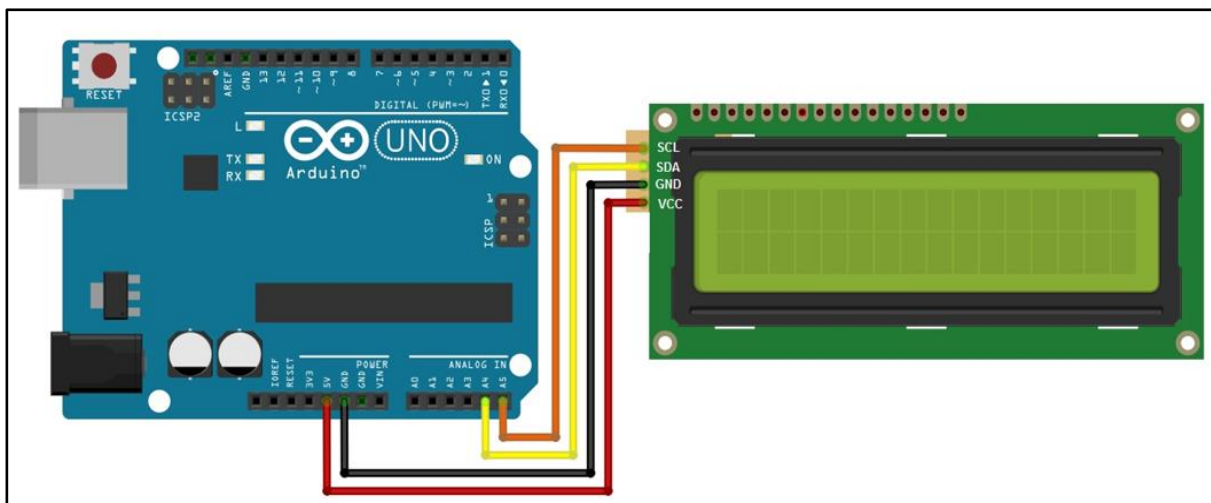
I2C LCD uses I2C communication interface to transfer the information required to display the content. I2C LCD requires only 2 lines (SDA and SCL) for transferring the data. So, the complexity of the circuit is reduced.



I2C Display

Interfacing LCD I2C Display with Arduino UNO:

I2C Connector	Arduino UNO
GND	GND
VCC	5V
SDA	A4
SCL	A5



8.0 Program Code:

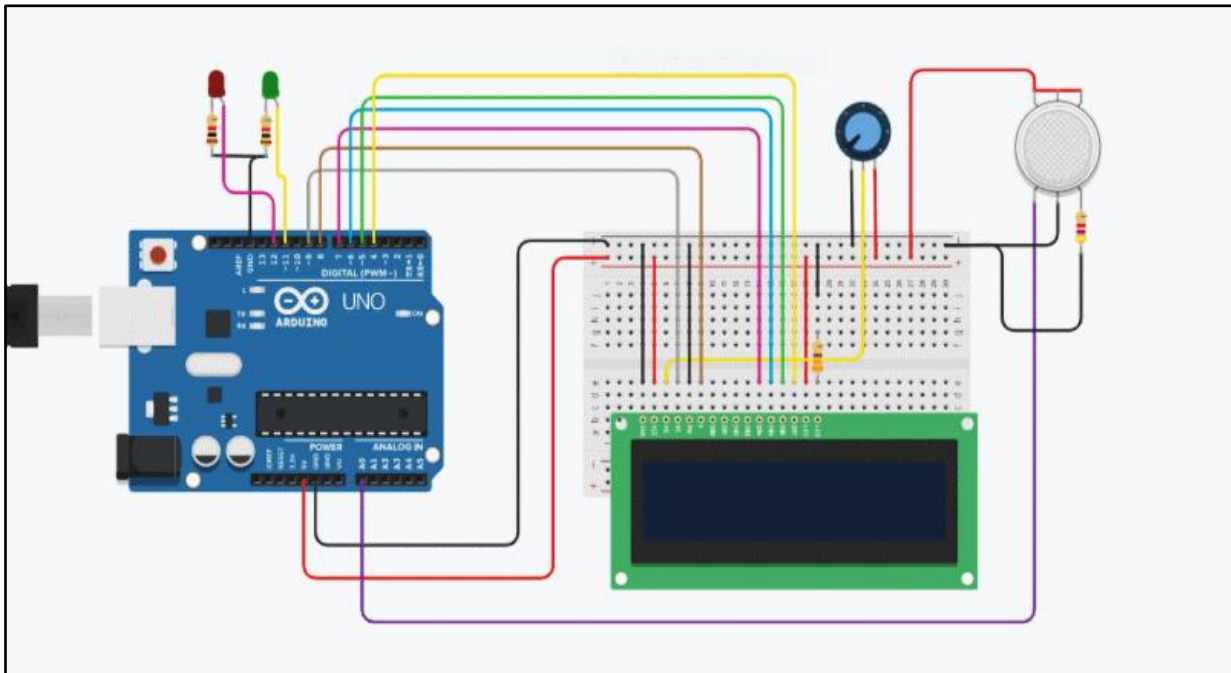
```
#include <LiquidCrystal_I2C.h>
#include <Wire.h> //For I2C Communication

int GreenLedPin = 6;
int RedLedPin = 7;
int MQ2SensorPin = 5; //Digital Pin

const int i2cAddress = 0x27; // Address of LCD module
const int lcdColumns = 16; //LCD columns
const int lcdRows = 2; //LCD row

//LCD object with the I2C address, number of columns, and number of rows
LiquidCrystal_I2C lcd(i2cAddress, lcdColumns, lcdRows);
void setup() {
  lcd.init();
  lcd.backlight();
  pinMode(GreenLedPin, OUTPUT);
  pinMode(RedLedPin, OUTPUT);
  pinMode(MQ2SensorPin, INPUT);
  Serial.begin(9600); // Initialize serial communication
}
void loop() {
  int sensorValue = digitalRead(MQ2SensorPin);
  if (sensorValue == LOW)
  {
    digitalWrite(RedLedPin, HIGH);
    digitalWrite(GreenLedPin, LOW);
    lcd.clear(); // Clear the LCD screen
    lcd.setCursor(0, 0); // Set cursor to the beginning of the first row
    lcd.print("Gas Detected!"); // Display message on LCD
    Serial.println("Gas Detected!"); // Print message on serial monitor
  }
  else
  {
    digitalWrite(GreenLedPin, HIGH);
    digitalWrite(RedLedPin, LOW);
    lcd.clear(); // Clear the LCD screen
    lcd.setCursor(0, 0); // Set cursor to the beginning of the first row
    lcd.print("No Gas Detected"); // Display message on LCD
    Serial.println("No Gas Detected"); // Print message on serial monitor
  }
  delay(1000);
}
```

Connections and interfacing of Components:



9.0 Skill Developed / Learning outcome of this Mini-Project:

9.1 Soft Skills:

- Problem Solving
- Leadership
- Teamwork
- Time Management
- Decision Making

9.2 Technical Skills:

- **Arduino programming:** Writing code to interact with sensors and control peripheral devices.
- **Sensor interfacing:** Understanding and implementing communication protocols to interface with the MQ2 sensor.
- **Circuit design:** Designing and assembling electronic circuits for sensor integration.

10.0 Applications of Mini Project:

Residential homes: Enhancing safety by detecting gas leaks from appliances such as stoves and heaters.

Commercial buildings: Protecting occupants and assets by monitoring gas levels in kitchens, laboratories, and storage areas.

Industrial facilities: Preventing accidents and ensuring worker safety by monitoring gas levels in manufacturing and processing plants.

10.0 Name of Group Members:

Roll No.	Name of Students	Student Signature
66	Mohan Suresh Ugale.	
51	Jayesh Bhikaji Pendharkar.	
30	Prathamesh Kiran Khule.	
09	Shivam Annasaheb Chavan.	

Date: 30 / 04/2024

Evaluated by: Dated Signature of Guide: _____

Name of Guide: Prof. Mrs. M.P. Joshi